Lab Experiment - 2

Aim

To implement a basic Finite State Automaton (FSA) in Python that recognizes strings ending with ab.

Algorithm

1. Start the program.
2. Read a string from the user.
3. Initialize the automaton state as q0.
4. Process each character of the string:
    - If the character is a, move to state q1.
    - If the character is b, move to state q2 when the previous character was a.
    - Otherwise, update the state based on the current character.
5. Check whether the final state is q2.
6. If the final state is q2, print Accepted; otherwise, print Rejected.
7. Stop the program.

Code

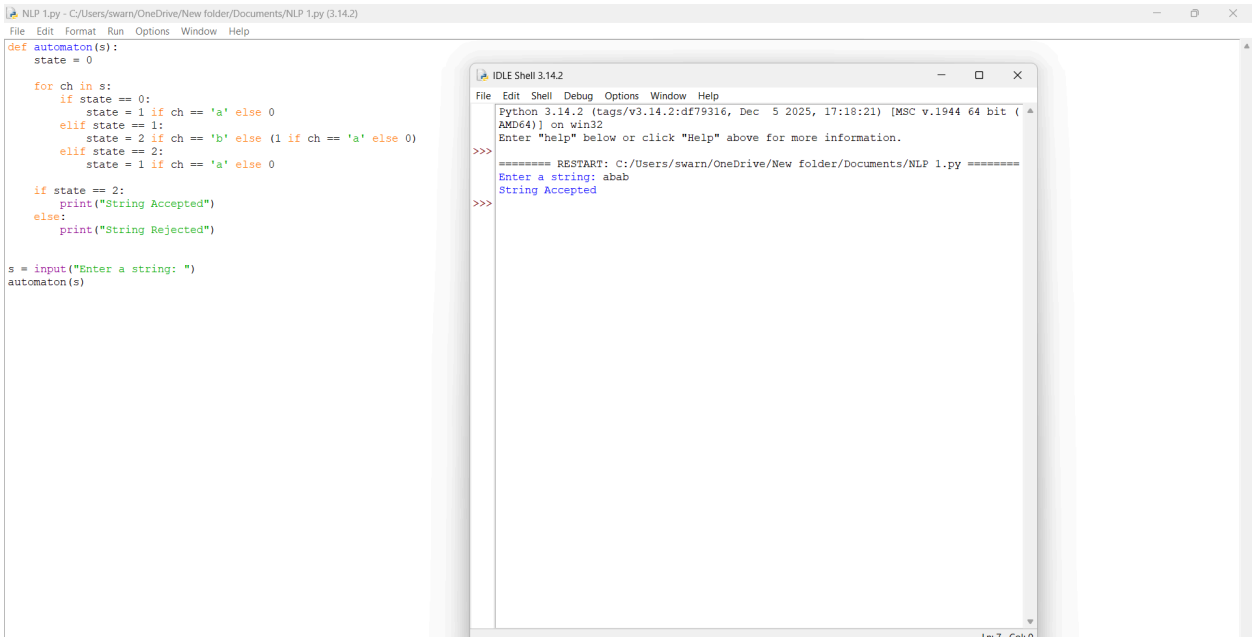
```
def automaton(s):
    state = 0

    for ch in s:
        if state == 0:
            state = 1 if ch == 'a' else 0
        elif state == 1:
            state = 2 if ch == 'b' else (1 if ch == 'a' else 0)
        elif state == 2:
            state = 1 if ch == 'a' else 0

    if state == 2:
        print("String Accepted")
    else:
        print("String Rejected")


s = input("Enter a string: ")
```

automaton(s)



Result

Thus, the Finite State Automaton was implemented successfully in Python to recognize strings ending with ab. Strings ending with ab are accepted, while other strings are rejected